

DS360 - Project Phase 2

WAN Link, Static Routing, and Testing

1. Project Overview

The purpose of this phase is to establish communication between two WAN sides through two routers. **Router0** represents WAN 1 and **Router1** represents WAN 2. The routers are connected by a serial point-to-point network. Static routes are then configured so that each router knows how to reach the remote LAN networks. Finally, connectivity is verified using ping tests in both directions.

Required WAN network: 10.10.10.0/30. **Router0 serial IP:** 10.10.10.1/30. **Router1 serial IP:** 10.10.10.2/30.

2. Addressing and Routing Plan

Device / Network	IP Address / Network	Subnet Mask	Purpose
Router0 Serial0/1/0	10.10.10.1	255.255.255.252 (/30)	WAN serial link
Router1 Serial0/1/0	10.10.10.2	255.255.255.252 (/30)	WAN serial link
WAN 1 - LAN 1	172.16.10.0/24	255.255.255.0	Local to Router0
WAN 1 - LAN 2	172.16.20.0/24	255.255.255.0	Local to Router0
WAN 2 - LAN 3	172.16.30.0/24	255.255.255.0	Remote from Router0
WAN 2 - LAN 4	172.16.40.0/24	255.255.255.0	Remote from Router0

3. Step 1 - Configure the Serial WAN Link

Connect Router0 and Router1 using the serial interfaces specified in the project. The serial network uses a /30 subnet because only two usable IP addresses are needed for a point-to-point link.

3.1 Router0 Serial Configuration

```
enable
configure terminal
interface serial0/1/0
ip address 10.10.10.1 255.255.255.252
clock rate 64000
no shutdown
exit
```

Explanation: The interface is assigned 10.10.10.1/30. The clock rate command is used only on the DCE side of the serial connection. The no shutdown command activates the interface.

3.2 Router1 Serial Configuration

```
enable
configure terminal
interface serial0/1/0
ip address 10.10.10.2 255.255.255.252
no shutdown
```

```
exit
```

Explanation: Router1 receives the second usable address in the /30 WAN network. If Router1 is the DCE side instead of Router0, the clock rate command should be configured on Router1 rather than Router0.

4. Step 2 - Configure Static Routing

A static route tells a router where to send packets for a network that is not directly connected. Router0 must know how to reach the two LANs behind Router1, while Router1 must know how to reach the two LANs behind Router0.

4.1 Static Routes on Router0

```
configure terminal
ip route 172.16.30.0 255.255.255.0 10.10.10.2
ip route 172.16.40.0 255.255.255.0 10.10.10.2
end
```

Meaning: For traffic going to 172.16.30.0/24 or 172.16.40.0/24, Router0 forwards the packet to the next-hop address 10.10.10.2, which is Router1.

4.2 Static Routes on Router1

```
configure terminal
ip route 172.16.10.0 255.255.255.0 10.10.10.1
ip route 172.16.20.0 255.255.255.0 10.10.10.1
end
```

Meaning: For traffic going to 172.16.10.0/24 or 172.16.20.0/24, Router1 forwards the packet to 10.10.10.1, which is Router0.

5. Step 3 - Verify Router Interfaces

Before testing PCs, verify that the router interfaces are active and have the correct addresses.

```
show ip interface brief
```

Expected check: Serial0/1/0 on each router should show the correct IP address and should normally be in an up/up state. If the interface is administratively down, enter interface configuration mode and use no shutdown. If the line protocol is down, check the cable, DCE clock rate, and the serial settings on both ends.

6. Step 4 - Verify the Routing Tables

```
show ip route
```

Router0 should contain static routes to: 172.16.30.0/24 and 172.16.40.0/24 through 10.10.10.2.

Router1 should contain static routes to: 172.16.10.0/24 and 172.16.20.0/24 through 10.10.10.1.

In the routing table, static routes are usually marked with the letter S. The WAN network 10.10.10.0/30 should appear as directly connected when the serial interfaces are correctly configured.

7. Step 5 - End-to-End Connectivity Testing

After the serial link and static routes are working, test communication between a PC in WAN 1 and a PC in WAN 2. The project document gives example destination addresses 172.16.30.2 and 172.16.10.2.

7.1 Ping from WAN 1 to WAN 2

On a WAN 1 PC, open Desktop > Command Prompt and run:

```
ping 172.16.30.2
```

Required successful result: Sent = 4, Received = 4, Lost = 0 (0% loss).

7.2 Ping from WAN 2 to WAN 1

On a WAN 2 PC, open Desktop > Command Prompt and run:

```
ping 172.16.10.2
```

Required successful result: Sent = 4, Received = 4, Lost = 0 (0% loss).

8. Troubleshooting if the Ping Fails

- Check the IP address, subnet mask, and default gateway on both PCs.
- Check that Router0 and Router1 serial interfaces use 10.10.10.1/30 and 10.10.10.2/30.
- Use show ip interface brief and make sure the serial interfaces are up/up.
- Confirm that the clock rate is configured on the DCE side of the serial cable if required.
- Confirm that no shutdown has been entered on both serial interfaces.
- Use show ip route and verify that all four remote-network static routes appear correctly.
- Check that the next-hop address on Router0 is 10.10.10.2 and on Router1 is 10.10.10.1.
- Test the WAN link directly by pinging 10.10.10.2 from Router0 and 10.10.10.1 from Router1 before testing end devices.

9. Complete Configuration Commands

Router0

```
enable
configure terminal
interface serial0/1/0
ip address 10.10.10.1 255.255.255.252
clock rate 64000
no shutdown
exit
ip route 172.16.30.0 255.255.255.0 10.10.10.2
ip route 172.16.40.0 255.255.255.0 10.10.10.2
end
show ip interface brief
show ip route
```

Router1

```
enable
configure terminal
```

```
interface serial0/1/0
ip address 10.10.10.2 255.255.255.252
no shutdown
exit
ip route 172.16.10.0 255.255.255.0 10.10.10.1
ip route 172.16.20.0 255.255.255.0 10.10.10.1
end
show ip interface brief
show ip route
```

10. Evidence Required for Submission

1. A screenshot showing the serial link between Router0 and Router1.
2. Screenshot(s) showing the serial IP configuration and the static route commands.
3. A screenshot of a successful ping from WAN 1 to WAN 2 showing 0% packet loss.
4. A screenshot of a successful ping from WAN 2 to WAN 1 showing 0% packet loss.

11. Suggested GitHub Commit Messages

Use commit messages that describe the real work completed. Appropriate examples are:

- Configure serial WAN connection
- Add static routing between WANs
- Verify end-to-end connectivity

Important: Upload the actual Packet Tracer file and real evidence created from your work. Do not create backdated or fabricated project history.

12. Final Written Conclusion

Conclusion: This phase configures the WAN connection between Router0 and Router1 using the 10.10.10.0/30 serial network. Router0 uses 10.10.10.1/30 and Router1 uses 10.10.10.2/30. Static routes are added so that Router0 can reach networks 172.16.30.0/24 and 172.16.40.0/24, while Router1 can reach networks 172.16.10.0/24 and 172.16.20.0/24. After the interfaces and routing tables are verified, ping tests must be performed in both directions. A correct configuration should produce four successful replies with 0% packet loss. The required screenshots should then be included as evidence of the completed work.